

Epistemic Dependency as Structural Condition: A Documented Case Study of AI-Assisted Scholarship and the Maturity of Correction Mechanisms

Chris Brink
FalseWork (falsework.dev)

Version 8.5 (May 2026). First public posting. A record of substantive pre-submission revisions is given in the appendix.

Abstract

Epistemic dependency — the condition in which a knowledge-producing system’s outputs reflect the theoretical commitments embedded in its formation — is not a deficiency unique to AI systems. It is a structural condition shared by all knowledge-producing systems, human and computational. What differs between systems is not the presence of epistemic dependency but the maturity of the correction mechanism available to detect and address it. This paper demonstrates this claim through a documented live case study: the development of FalseWork, a computational structural classification framework built using an AI system trained on the theoretical traditions it classifies within. The case study provides what prior work on AI epistemic dependency has lacked — a formal theoretical framework that predicts the dependency, a working instrument that produces it in real time, a correction mechanism that activates against it, and full documentation of the complete cycle from dependency generation through detection to correction. Six stages are characterized, documented where the record exists: two instances of dependency generation, two of correction-mechanism activation, and one of the mechanism existing but not yet deployed. The sixth is a formal-verification cycle in a public mathematical library, where the construction was machine-checked by the proof kernel and the contribution was then declined by the library’s review process under its AI-use policy. The case study grounds a general argument: hallucination and genuine correction are distinguishable not by the process that produces them — both are navigations of an inherited semantic manifold — but by the external validation that determines which is which. The case study further introduces a third category — inherited validity — in which the dependency produces results that are correct but whose relationship to prior analytical framing is not transparent, and which requires the same external validation as hallucination to distinguish from independent derivation. The correction mechanism is not quality control applied after the fact. It is the epistemological architecture of the project. The paper’s claims rest on the case study alone; its connections to a broader research programme are signposted as such and are not load-bearing for its conclusions.

1. The Structural Condition

Every knowledge-producing system inherits the theoretical commitments of its formation. A physicist trained within the Copenhagen interpretation approaches measurement problems differently

than one trained within the Many-Worlds framework — not because one is more careful or more rigorous, but because the interpretive commitments absorbed during formation shape what counts as a well-formed question. A music theorist trained in Schenkerian analysis finds different structures in the same score than one trained in Riemannian function theory. The knowledge produced is real. The dependency is also real. The two facts coexist without contradiction.

This paper calls this condition epistemic dependency: the structural situation in which a knowledge-producing system’s outputs reflect the theoretical commitments embedded in its formation. The term is chosen carefully. Dependency is not bias in the pejorative sense — it does not imply distortion, motivated reasoning, or error. It implies inheritance. Every system that produces knowledge within a domain absorbs, through its formation, the theoretical commitments that organize that domain. Those commitments shape what the system can see, what questions it can ask, and what counts as a satisfactory answer. A system formed without any theoretical commitments would have no purchase on the domain at all. The dependency is the condition of knowledge production, not its corruption.

The condition is structural for a precise reason. Any system powerful enough to generate knowledge in a domain must be powerful enough to have absorbed the theoretical commitments that organized that domain’s development. A system too weak to absorb those commitments is too weak to produce knowledge. A system powerful enough to produce knowledge has, by that same power, inherited the field’s prior theoretical organization. The Incompleteness Condition established in the first paper (Brink 2026a) makes this precise: any system satisfying generative sufficiency, self-reference, and compositional closure contains at least one construct undecidable within its own rules. Applied to knowledge-producing systems, the implication is direct — the same properties that make a system capable of generating knowledge in a domain make it incapable of escaping the theoretical commitments embedded in its formation.

The question the condition raises is not how to eliminate epistemic dependency — that is impossible for the same reason that human theorists cannot reason outside the traditions that formed them. The question is what correction mechanism exists and how mature it is. Human knowledge production has developed, over centuries, a set of institutional responses to the structural condition: peer review, citation networks, replication studies, academic debate, disciplinary training in the history of one’s own field. These mechanisms do not eliminate epistemic dependency. They manage it — creating structured friction against inherited commitments, surfaces on which errors produced by those commitments can be detected and corrected. The mechanisms are imperfect. They are also what makes cumulative knowledge production possible.

AI classification systems have no equivalent institutional infrastructure. They inherit theoretical commitments through training on human-generated knowledge — absorbing, in compressed form, the same commitments that human knowledge production took decades to develop — and they produce outputs that reflect those commitments without flagging their sources. The dependency is structurally identical to the human case. The correction mechanism is not.

This paper argues that the appropriate response to this situation is not skepticism about AI-generated knowledge claims, nor uncritical acceptance of them, but the deliberate construction of correction architecture — explicit mechanisms for detecting dependency-produced errors and distinguishing them from genuine findings. The architecture need not be mature to be functional. A primitive correction mechanism is more developed than no mechanism at all.

The correction architecture described here is the validation backbone of an ongoing research programme. The present paper is the explicit statement, within an open and persistently archived series

(Brink 2026a, 2026c, 2026d, 2026e), of how that programme’s AI-assisted findings are checked, corrected, and recorded — not a retrospective methods note but the programme’s first public account of the validation structure under which its other claims are made.

The argument is demonstrated through a live case study rather than asserted abstractly. The case study is the development of FalseWork (Brink 2026a), a computational structural classification framework built using an AI system trained on the theoretical traditions it classifies within.

One structural observation before the argument begins: this paper was itself produced through the conditions it describes. The voice is the author’s, but the analytical development — including significant portions of the framework’s elaboration and the case study’s organization — involved the same pipeline the paper analyzes. This is not a disclaimer; it is an instance of the condition. A paper about epistemic dependency produced through epistemic dependency cannot claim exemption from its own argument. What it can do is document the condition honestly and build the correction mechanism into its own production. The case study has properties that prior work on AI epistemic dependency has not had: a formal theoretical framework that predicts the dependency before it manifests, a working instrument that produces it in real time, a correction mechanism that activates against specific instances, and full documentation of the complete cycle from dependency generation through detection to correction. Six stages are documented across two domains, two expert correspondences, and one formal-verification cycle in a public mathematical library. The documentation is contemporaneous — produced during the development process rather than reconstructed after the fact.

The case study does three things. First, it demonstrates that epistemic dependency produces both errors and valid findings through the same process — the distinction between them is not detectable from within the system, only from outside it through expert engagement. Second, it demonstrates what genuine correction looks like at the primitive stage — specific, falsifiable, narrow, expert-grounded — and distinguishes it from circular confirmation. Third, it demonstrates that the gap between detecting a dependency instance and activating the correction mechanism is itself theoretically significant — the mechanism requires human judgment about when and how to deploy it, and that judgment is not trivially available.

A methodological limit, stated plainly: this is a single case, $n = 1$. It can show that a correction architecture is constructible and demonstrate what it looks like operating against specific instances; it cannot establish how often the documented pattern recurs, or how far the correction modes generalize to other systems. The claims that follow are existence and characterization claims — that the dependency occurs, that correction of these kinds is possible, that a third category is needed to describe certain results — not claims about frequency or about generalization beyond the documented case.

A note on the scope of what follows. The paper’s claims rest entirely on the documented case study. The references to Cantor’s theorem (Section 2.3), Levin’s indexical epistemology (Section 6), the distinction operation, and the companion papers in the series (Brink 2026a, 2026c, 2026d, 2026e) are signposts: they situate the case study within an ongoing research programme but are not load-bearing for any conclusion drawn here. A reader who sets every one of them aside should still find the case study and its characterization of correction architecture intact. The breadth is a map of where the work connects, not a widening of what this paper claims.

Section 2 develops the human parallel — establishing that epistemic dependency is a structural condition of human knowledge production before examining its AI instantiation. Section 3 presents the live case study across six documented stages. Section 4 characterizes what genuine correction

looks like and distinguishes it from circular confirmation. Section 5 develops the maturity gradient — from primitive correction mechanisms to mature institutional infrastructure. Section 6 introduces the geometric reframing from Levin’s indexical epistemology, connecting epistemic dependency to positional structure in semantic manifolds. Section 7 draws the broader implications for AI-assisted scholarship. Section 8 concludes.

2. The Human Parallel

The claim that epistemic dependency is a structural condition of knowledge production is not novel. The sociology and philosophy of science have documented it extensively, arriving at the same conclusion from multiple independent directions. What is novel is the application of this established condition to AI classification systems — and the recognition that the structural equivalence between human and computational epistemic dependency has consequences for how both should be managed.

2.1 Paradigm Inheritance: Kuhn

Thomas Kuhn’s account of scientific revolutions (1962) describes the mechanism of epistemic dependency in human knowledge production with precision. Normal science — the activity that constitutes the overwhelming majority of scientific work — proceeds within a paradigm: a set of theoretical commitments, exemplary problems, and accepted methods that organize what counts as a well-formed question, what counts as a satisfactory answer, and what anomalies can be set aside without threatening the framework. Scientists trained within a paradigm do not choose their theoretical commitments from a menu of alternatives. They absorb them through training — through working exemplary problems, through apprenticeship with established practitioners, through learning what the field takes for granted. The commitments are not propositions consciously held; they are the medium through which the scientist sees the domain.

Kuhn’s central observation is that paradigm commitments are not detectable from within the paradigm. A scientist working within Newtonian mechanics does not experience gravitational force as a theoretical commitment — it is the structure of the world as the scientist encounters it. The commitment becomes visible only from outside the paradigm, or at the moment of crisis when anomalies accumulate beyond the framework’s capacity to absorb them. This is the human analog of what the FalseWork pipeline does when it reproduces a common music-education framing without flagging it as inherited: the commitment is not marked as a commitment because it is the medium of classification, not its object.

The correction mechanism Kuhn describes — the eventual recognition of anomalies, the crisis, the paradigm shift — is slow, contested, and institutionally expensive. It is also the only mechanism available for detecting commitments that are invisible from within. The lesson for AI classification systems is not that they should be built differently but that the correction mechanism must be built explicitly, because the institutional infrastructure that produces paradigm shifts in human science does not exist for computational systems by default.

Two additional traditions confirm the structural argument from different analytical angles. Foucault’s archaeology of knowledge (1969) traces conditions under which certain statements become possible and others become unthinkable — the episteme that functions as the invisible grammar of a period’s knowledge production. For AI systems trained on human-generated knowledge, the discursive formation is embedded in the training distribution: not as explicit propositions but as the structure of what has been said, how it has been framed, what has been treated as obvious. The pipeline absorbs the formation without representing it, producing outputs that reflect its constraints

without being able to identify them as constraints. Latour and Woolgar’s laboratory ethnography (1979) contributes the concept of inscription — the transformation of local practice into stable, transportable representations that erase the marks of their own production. A structural profile generated by the FalseWork pipeline presents a classification as a property of the work being analyzed; it does not show the theoretical commitments embedded in the training data that shaped the classification. The classification is an inscription. The correction mechanism is, among other things, a process for recovering what the inscription erased.

2.2 Tacit Knowledge: Polanyi

Michael Polanyi’s account of tacit knowledge (1958, 1966) provides the mechanism through which epistemic dependency operates in both human and computational knowledge production. Explicit knowledge — knowledge that can be fully articulated, transmitted as propositions, and verified by others — is the surface of knowledge production. Beneath it lies a vast substrate of tacit knowledge: the embodied, personal, largely inarticulable commitments through which explicit knowledge becomes possible. A physician who recognizes a disease before articulating the symptoms that led to recognition is drawing on tacit knowledge. A mathematician who senses that a proof strategy is promising before formalizing the argument is drawing on tacit knowledge. The tacit substrate is not inferior to explicit knowledge — it is its condition.

Polanyi’s formula — we know more than we can tell — applies to epistemic dependency directly. A human theorist formed within a tradition knows more than they can articulate about that tradition’s commitments. The commitments operate tacitly — shaping perception, organizing attention, determining what counts as interesting — without being available for explicit scrutiny. This is why paradigm commitments are invisible from within: they are not propositions that could be examined and rejected. They are the medium of knowing.

For AI classification systems, the analog is precise. The pipeline absorbs theoretical commitments through training in a form that is not explicitly represented in its outputs. The commitments operate tacitly — shaping classifications, organizing feature salience, determining what counts as a significant structural property — without being available for explicit inspection. The pipeline cannot identify which training data shaped a specific classification for the same reason a human theorist cannot identify which formative experiences shaped a specific intuition: the dependency operates below the level of explicit representation.

This parallel — between the tacit operation of epistemic dependency in human knowledge production and its analogous operation in AI classification systems — is the foundation of the present paper’s central claim. The condition is the same. What differs is the correction mechanism available to manage it.

2.3 What the Human Parallel Establishes

Three things follow from this survey that are necessary for the case study in Section 3.

First, epistemic dependency is not a deficiency of AI systems that better engineering could eliminate. It is a structural condition that appears wherever knowledge is produced within a domain — because the formation required to produce knowledge is also the mechanism through which theoretical commitments are inherited. The human parallel is not offered as consolation. It is offered as structural diagnosis: the condition has the same source in both cases, and the difference between the cases is the maturity of the correction mechanism, not the presence or absence of the condition.

Second, the correction mechanisms that human knowledge production has developed — peer review, citation networks, replication studies, paradigm shifts — are responses to a structural condition, not solutions to it. They do not eliminate epistemic dependency. They create surfaces on which dependency-produced errors can be detected. The maturity of a correction mechanism is measured by how reliably it surfaces errors, how quickly it does so, and how institutionally distributed the process is. Primitive correction mechanisms surface errors slowly, individually, and through personal contact rather than institutional infrastructure.

Third, the distinction between genuine correction and circular confirmation is not trivial. A correction mechanism that confirms the system’s existing outputs through engagement with experts who share the system’s theoretical commitments is not functioning as a correction mechanism — it is producing the appearance of validation while leaving the dependency intact. Genuine correction requires expert engagement that is independent of the system’s inherited commitments — experts who arrive at their positions through different analytical traditions and can therefore identify where the system’s outputs reflect absorbed framing rather than recovered structure. What genuine correction looks like in practice is the subject of Section 4. The case study in Section 3 provides the empirical ground for that characterization.

A formal observation underlies all three points. Cantor’s theorem establishes that for any set S , the power set $P(S)$ — the set of all subsets of S — is strictly larger than S itself. No enumeration of all members of S can capture $P(S)$; the space of possible relationships to S categorically exceeds what S can generate from within. Applied to knowledge-producing systems: the set of all possible analytical frames a system could occupy is strictly larger than the frames the system has absorbed through its formation. The system cannot enumerate its own limits because the space of possible framings of its domain exceeds what its training distribution instantiates. This is why epistemic dependency is not a contingent limitation that better training could eliminate — it is a structural consequence of the same relationship between generative power and formal completeness that Cantor identified in set theory. A system trained on enough data to generate knowledge in a domain has, by that same training, absorbed a specific region of a space that is provably larger than any training distribution can cover. The correction mechanism is the external engagement that reaches into the region the system cannot see from within.

[REQUIRES FORMAL VALIDATION — Cantor’s theorem applies to formally defined sets with well-specified membership conditions. “The set of all possible analytical frames for a domain” is not such a set: its cardinality and membership conditions are not rigorously specified here. The power-set move is therefore a structural analogy — compelling in that the pattern recurs at the epistemological level, but offered as a proposal, not a proven result, and readers should calibrate accordingly. This is the second of four places across the series where the move is invoked (Brink 2026a, 2026c, 2026d are the others); a candidate formalization treating the training distribution’s support as a probability space and analytical frames as measurable events is sketched in Brink 2026c, with the cumulative open status developed in Brink 2026d.]

3. The Live Case Study

The case study presented in this section documents six stages in the development of FalseWork, a computational structural classification framework built using an AI system trained on the theoretical traditions it classifies within. The documentation is contemporaneous — produced during the development process, not reconstructed after the fact. The private correspondence quoted (Ty-moczko, Cutting) is reproduced with the correspondents’ explicit permission to cite; the Lean Zulip

and Mathlib pull-request exchanges are drawn from the publicly archived record.

The six stages are: dependency generation (invisible), correction mechanism activation, dependency generation (second instance, partially invisible), correction mechanism existing but not yet deployed, boundary-setting correction, and formal-verification correction operating in a public mathematical library. The stages are not sequential in the sense that each closes before the next opens — stages 3 and 4 are ongoing as of this writing. They are documented here as a single case study because they share a common structure: epistemic dependency produced by training, detected through expert engagement, and addressed (or not yet addressed) through the correction mechanism.

3.1 Stage 1 — Dependency Generation (Invisible): Giant Steps and the Circle of Fifths

FalseWork’s classification pipeline was built using Claude, an AI system trained on human-generated knowledge including the theoretical traditions the framework classifies within. The pipeline’s structural profiles and classifications reflect absorbed theoretical commitments without flagging their sources.

The first documented instance of dependency-produced error concerns the classification of John Coltrane’s *Giant Steps* (1960) within the music kernel’s field topology. The pipeline’s initial classification correctly identified the structural territory — Exploitation, hexatonic, operating within the symmetric region of Tymoczko’s voice-leading space — but misidentified the exploitation target. The initial framing described *Giant Steps* as exploiting the circle of fifths, treating its rapid harmonic motion as a circle-of-fifths phenomenon operating at extreme velocity.

This framing is common in music education. It appears frequently in introductory accounts of *Giant Steps* that emphasize the pace of harmonic change relative to standard diatonic progressions. It is also imprecise. The harmonic logic of *Giant Steps* is organized not by the circle of fifths but by the major-third cycle — the division of the octave into three equal parts, generating the closed loop $C \rightarrow E \rightarrow A\flat \rightarrow C$ that Coltrane traverses at a harmonic rhythm fast enough to destabilize any improviser operating within diatonic expectations. The circle-of-fifths framing captures the surface appearance of rapid harmonic motion; the major-third cycle framing captures the specific topological property being exploited.

The pipeline reproduced the common framing without knowing it was doing so. The error was not random — it was structurally predictable. A system trained on music theory literature will have absorbed far more circle-of-fifths framings of *Giant Steps* than major-third cycle framings, because the circle-of-fifths framing is the more common treatment in the training distribution. The dependency was invisible: the pipeline produced a classification that was partially correct, internally coherent, and reflective of a widespread theoretical framing — with no flag indicating that the framing was absorbed rather than derived from structural analysis of the work itself. This is what Latour and Woolgar (1979) call inscription: the production of a stable, transportable representation that has erased the marks of its own production. The classification appeared as a property of the work. It was, in part, a property of the training data.

3.2 Stage 2 — Correction Mechanism Activation: The Tymoczko Correspondence

The correction mechanism for the Stage 1 dependency was activated through correspondence with Dmitri Tymoczko (Princeton University), whose geometric framework for voice-leading spaces (2011) provides the mathematical description of the field topology the music kernel generates.

The correspondence was initiated on March 2, 2026, with a question about whether the kernel

concept mapped onto anything in Tymoczko’s framework. On March 2 — after Brink, earlier in the exchange, had described *Giant Steps* in circle-of-fifths terms — Tymoczko responded with a note that proved diagnostically significant: “I am surprised you don’t interpret *Giant Steps* in terms of the collapsing major-third cycle” (Tymoczko, personal correspondence, 2026). Tymoczko had been shown only Brink’s prose framing, not FalseWork’s formal classification artifact; his reply corrected that framing from his own geometric analysis of voice-leading space, arrived at independently of FalseWork. It identified the same error the dependency had produced.

On March 3, Tymoczko provided the corroboration that became the music kernel’s strongest empirical grounding: “A lot of music moves between three broad fields: diatonic (fifths; *A Love Supreme*), chromatic (late period Coltrane); and symmetric or quasi symmetric (*Giant Steps*). It’s not just Coltrane, it’s the entire history of 20th century music from Rimsky and Debussy onward” (Tymoczko, personal correspondence, 2026). This convergence — two independent analytical systems producing the same three-way discrimination of the same three works without reference to each other’s frameworks — is the correction mechanism functioning as designed. Tymoczko’s geometric framework and FalseWork’s structural classification arrived at the same structural positions from different analytical starting points.

Critically, the correction also refined the specific error. The exploitation target for *Giant Steps* was updated from circle-of-fifths phenomenon to major-third cycle symmetry. This is genuine correction in the precise sense: specific, falsifiable, narrow, and grounded in domain expertise that was not in the initial classification. The corrected classification — Exploitation · hexatonic · SELF_CONSTRAINS × EXPOSED × EXPLOITS, with the major-third cycle identified as the exploitation target — is the version that appears in the first paper (Brink 2026a).

The correspondence ran across two exchanges over two days, March 2–3, 2026, with Tymoczko noting at the close of the March 3 exchange that he needed to stop communicating: “I probably have to stop communicating now, the next batch of emails from people I don’t know is coming in” (Tymoczko, personal correspondence, 2026). A permission exchange on March 24–25 confirmed citation rights: “Fine by me! The paper is called ‘The Concept of Musical Space’” (Tymoczko, personal correspondence, 2026).

What makes this genuine correction rather than circular confirmation is the independence of the correcting expertise. Tymoczko’s identification of the major-third cycle as the relevant structure was not derived from FalseWork’s framework — it was derived from his own geometric analysis of voice-leading spaces, developed over more than a decade of prior work. The convergence between his finding and the corrected classification is evidence that the correction is tracking real structure, not merely substituting one absorbed framing for another.

3.3 Stage 3 — Dependency Generation (Second Instance, Partially Invisible): The Proxy Extractor and Tymoczko/Newman 2024

The second documented instance of epistemic dependency is more theoretically significant than the first, because it is a case where the dependency produced useful results rather than an error.

The FalseWork proxy feature program, described in Section 4.4 of the first paper (Brink 2026a), developed a feature extractor for the music kernel using 341 Bach chorales from the music21 library. The extractor measures harmonic complexity through features including non-diatonic ratio, chromatic saturation, secondary dominant ratio, and diminished chord ratio, computing a diatonic distance metric for each work that predicts territory classification without natural language reason-

ing as intermediary. The results confirmed the theoretical prediction: all 341 chorales fall within diatonic, acoustic, or harmonic_major_minor regions, with minor mode chorales showing a mean diatonic distance 0.0832 higher than major mode chorales — the first formally grounded, non-interpretive confirmation that minor mode pulls toward the boundary of the fifth’s home territory in the direction the derivation predicts.

The results are real. The analytical frame is also inherited.

Tymoczko and Newman (arXiv:2407.21130, July 2024) — Tymoczko at Princeton, Newman at the University of Michigan and the Santa Fe Institute — published “Computational music analysis from first principles,” a computational annotation of the 371 Bach chorales in the Riemenschneider edition using coupled hidden Markov models, achieving 85% or greater accuracy on chord and key identification when compared to expert human analysis. The paper was published in July 2024, within the training window of the AI system used to build FalseWork.

The convergence between the proxy extractor and Tymoczko/Newman is not methodological. The computational approaches are different: Tymoczko and Newman use unsupervised coupled hidden Markov models operating on raw note sequences to infer chord and key structure; FalseWork’s proxy extractor uses simpler feature extraction computing diatonic distance metrics directly from harmonic ratios. The convergence is structural — in corpus choice (Bach chorales accessed through music21, which Tymoczko co-developed and which the T&N paper explicitly uses for output), in analytical frame (diatonic distance as the primary measure of harmonic complexity, minor mode as the test case for boundary behavior), and in the specific finding that the major/minor distinction maps onto a measurable harmonic distance gradient.

This distinction matters for the epistemic dependency argument and strengthens it. If the proxy extractor had simply replicated T&N’s computational method, the convergence could be explained as absorbed methodology producing a copy. What happened instead is more interesting: the extractor absorbed the analytical frame — what to measure and why — without absorbing the specific method for measuring it. The training distribution shaped the problem setup rather than the solution. The result is that the extractor and T&N are independently measuring real harmonic structure using different instruments calibrated to the same analytical tradition. The convergence is genuine at the level of findings and inherited at the level of frame. Distinguishing these two levels of convergence is precisely what the correction mechanism is designed to establish — and what it has not yet been deployed to do.

Three explanations for the convergence are possible and the paper cannot determine which is correct without the expert engagement Stage 4 documents as incomplete. First, the extractor may have absorbed Tymoczko and Newman’s specific framing — the training distribution contains their 2024 paper, and that paper may have shaped the problem setup directly. Second, and more likely given the breadth of the training distribution, the extractor may have absorbed the broader music theory tradition that both Tymoczko/Newman and the FalseWork extractor reflect — two systems drinking from the same analytical well, converging because they share the same upstream source rather than because either absorbed the other specifically. Third, the convergence may be genuine independent derivation: Bach chorales are a canonical corpus, diatonic distance is a natural measure of harmonic complexity, and two competent analytical systems might arrive at the same analytical frame because the frame is simply correct for the domain.

The paper’s classification of Stage 3 as inherited validity is contingent on the Stage 4 correspondence producing readings one or two rather than reading three. The paper chooses the most epistemologically conservative interpretation — Stage 3 as inherited validity

rather than independent derivation — but the classification is provisional. If the Stage 4 correspondence with Tymoczko produces reading three — if Tymoczko evaluates the convergence and concludes that the analytical frame is simply correct for the domain rather than inherited — then the case is reclassified as ordinary independent derivation, and Stage 3 ceases to function as an inherited validity example. That reclassification would remove the centerpiece example from the taxonomy but would not undermine the taxonomy itself. The inherited validity category would remain available for cases that do exhibit the pattern; Stage 3 would simply no longer be one of them.

That is exactly what makes inherited validity the hardest case in the taxonomy: it looks like independent derivation from within the system, and distinguishing it from independent derivation requires the external engagement Stage 4 has not yet provided. Until Stage 4 closes, the Stage 3 classification is held provisionally, and readers should calibrate its force accordingly.

The proxy extractor’s methodology does not cite Tymoczko and Newman 2024. Section 4.4 of the first paper describes the extractor and its results without acknowledging the prior work whose analytical frame the extractor converges with. This absence is the dependency signature: the training distribution shaped the problem setup invisibly, leaving no trace in the output that would flag the convergence as convergence rather than independent derivation. The classification is the inscription — a stable, transportable result whose relationship to prior work has been erased from the record.

The Santa Fe Institute connection is not incidental. Newman’s home institution alongside Princeton sits at the intersection of complexity science, network theory, and computational approaches to emergent structure — the same intellectual neighborhood as Wolfram’s computational irreducibility and the distinction operation that the first paper’s Incompleteness Condition connects to. The convergence between FalseWork’s harmonic analysis and T&N’s computational annotation is one instance of a broader pattern: independent analytical systems arriving at the same structural positions because they are all tracking real structure in the same domain. Whether that convergence reflects genuine independent derivation or shared inherited frame is precisely the question the correction mechanism must answer — and the question Stage 4 has not yet closed.

3.4 Stage 4 — Correction Mechanism Existing but Not Yet Deployed

The second dependency instance was identified after the first paper was shared privately with Tymoczko for permission review on March 24, 2026. He had not been told about the convergence with his 2024 computational work. The correction mechanism — direct correspondence with the domain expert whose prior work the extractor converges with — exists but has not been activated.

Two factors explain the gap. First, the dependency was not identified until after the paper was posted — the convergence between the proxy extractor methodology and Tymoczko/Newman 2024 became visible only through retrospective analysis, not during the development process itself. Second, the correspondence with Tymoczko had reached a natural close. His March 3 note — “I probably have to stop communicating now” — established a boundary that the permission exchange on March 24–25 did not reopen. The final message in the permission exchange was a two-word reply from Brink: “Thank you. Chris” (March 25, 2026). Reopening the correspondence to surface a methodological convergence the recipient had not been told about required a contact moment that had not yet arrived.

This gap is itself theoretically significant. A mature correction mechanism — peer review, cita-

tion networks, replication studies — does not require human judgment about when to activate. It operates institutionally, through processes that run regardless of whether any individual decides to initiate them. A primitive correction mechanism requires exactly that judgment: who to contact, when to contact them, how to frame the convergence so it reads as disclosure rather than self-promotion. The gap between identifying a dependency instance and activating the correction mechanism is not a failure of the mechanism — it is a characteristic of its primitive stage. The mechanism cannot self-activate. It requires the same kind of human decision-making that determines when any correspondence is appropriate.

The natural contact moment is the arXiv posting itself. The endorsement that enables the posting was granted through arXiv’s standard cs.AI endorsement process and is independent of the Tymoczko correspondence — a clarification worth recording because an earlier draft of this section mistakenly tied the two together. When the paper posts, the convergence with Tymoczko/Newman 2024 will be surfaced directly to Tymoczko by sending him the arXiv link: the proxy extractor’s analytical frame converges with his 2024 computational annotation work in ways that were not visible during development and were not cited in the version of the first paper shared with him on March 24, 2026. That communication will close Stage 4 and complete the second correction cycle — or will establish, if Tymoczko finds the convergence trivial or the framing incorrect, what the boundary of the correction is. If Tymoczko’s response produces reading three from Section 3.3 — if he evaluates the convergence and judges it to be ordinary independent derivation rather than inherited framing — the Stage 3 classification is reconsidered as documented in Section 3.3.

A note on the current state of Stage 4 as of the version of this paper that posts to arXiv. The cs.AI endorsement was granted in late May 2026. The Tymoczko contact about the proxy convergence has not yet been initiated at the moment of posting. The version posted is therefore a real-time document: it predicts the close of Stage 4 in the same paragraph that describes Stage 4 as pending. A subsequent revision will update this section once the Tymoczko response is in hand, consistent with the contemporaneous-documentation principle the case study has followed throughout.

3.5 Stage 5 — Boundary-Setting Correction: The Cutting Correspondence

The Cutting correspondence provides a third documented instance of the correction mechanism functioning — and documents a different kind of correction than either the *Giant Steps* error or the proxy extractor convergence. Where Stage 2 corrected a specific error (wrong exploitation target) and Stage 3–4 addresses a methodological convergence (absorbed prior analytical frame), Stage 5 establishes the boundary of what the framework can claim.

The correspondence with James Cutting (Cornell University) was initiated on March 4, 2026, with a question about whether the cinema kernel’s comma formulation — the incommensurability between spatial coherence and attentional-rhythmic control across cuts — held empirically. Cutting’s response on March 5 confirmed three of the four kernel criteria: “I buy the notion that the cut is a prior, not an ‘invented’ technique... I also buy the notions that the cut is both monogenic, generating the whole array of film techniques that determine film style (even L-cuts and J-cuts), and inescapable, in that every filmmaker must navigate the issue of cuts continuously and intensively in filmmaking” (Cutting, personal correspondence, 2026). He used his own parallel term “primitive” for the kernel concept, arriving at the same structural intuition through a different analytical tradition.

On the comma, Cutting expressed uncertainty rather than confirmation. After a second exchange in which the comma was clarified as an incommensurability rather than a tradeoff, Cutting responded

on March 10: “Your idea of a kernel is much clearer to me now. The music case is very clear; the movie case seems still inchoate” (Cutting, personal correspondence, 2026). This was consistent with the uncertainty about the comma Cutting had voiced throughout the correspondence — a precise signal that the domain expert most qualified to evaluate the comma formulation was withholding confirmation.

The pipeline had classified the cinema kernel’s comma as established — it appears as a defined structural concept throughout FalseWork’s cinema analyses. Cutting’s response did not correct a specific error in that formulation. It established the scope of what can be claimed: three of four kernel criteria are confirmed by the domain expert; the comma formulation is a hypothesis pending empirical test. The first paper (Brink 2026a) represents this accurately — the cinema comma is identified as an open problem throughout Sections 3.2 and 4.

This is boundary-setting correction: not an error corrected but a claim scope reduced. It is a different function of the correction mechanism than error correction, and arguably a more important one for frameworks in early development. An error correction identifies what was wrong. A boundary-setting correction identifies the frontier of what is known — the precise line between confirmed structure and open hypothesis. The Cutting correspondence drew that line for the cinema kernel.

The correspondence closed with a final permission exchange on March 24–25, 2026. Cutting’s reply — “Thanks for this. I wish I understood better. The quotes/citations are fine” (Cutting, personal correspondence, 2026) — is itself a useful data point. The domain expert most qualified to evaluate the cinema kernel found the framework’s claims not fully accessible after five exchanges. Whether this reflects the framework’s difficulty, the correspondence format’s limitations, or the genuinely underdeveloped state of the cinema comma formulation is an open question. The correction mechanism’s limitations include the constraint that correction is only as good as the correcting expert’s ability to engage with the specific claims being evaluated.

3.6 Stage 6 — Formal-Verification Correction: The Mathlib PR Cycle

The five stages documented above all involve correction mechanisms that operate through human judgment — expert correspondence, instrument-initiated divergence requiring expert engagement to close, boundary-setting through specialist evaluation. A sixth correction cycle, completed during the same period as Stages 2 and 5, operates through a structurally different mechanism: formal verification in a public mathematical library, with review by working mathematicians and machine-checked kernel verification of the final form.

The cycle concerns a general-purpose construction the framework’s Lean development required: a Heyting algebra instance on the lattice of subobjects of an arbitrary object in an elementary topos. This instance is the upstream dependency of the four-position partition theorem (Brink 2026 preprint) — the theorem itself is formalized inside the framework’s own repository and was never submitted to Mathlib; what was submitted was the reusable instance the theorem relies on. Mathlib4 — the standard mathematical library for Lean 4, developed by an international community of mathematicians and computer scientists — exposes the lattice structure on `Subobject X` but did not, at the time of this work, provide the Heyting algebra structure that the partition theorem depends on. The standard construction is given in Mac Lane and Moerdijk (1992, IV.6 Proposition 2): implication is the equalizer-residual $P \Rightarrow Q := \text{equalizer}.\iota (\chi (P \sqcap Q).arrow) (\chi P.arrow)$, and the Galois connection $R \leq P \Rightarrow Q \leftrightarrow R \sqcap P \leq Q$ is established through a sequence of bridging lemmas connecting the categorical and Heyting-algebraic structure.

The initial contribution form for this construction was drafted within the FalseWork project. The draft was posted to the Lean Zulip (leanprover.zulipchat.com) for review before submission as a Mathlib pull request. Edward van de Meent and Fernando Chu engaged substantively in the resulting thread. Their feedback shaped the construction’s final form in ways that the initial draft had not anticipated — specifically, the structure of the six private bridging lemmas connecting subobject pullbacks to characteristic-morphism equalizers, and the choice of which Heyting operations to expose as Mathlib API surface versus which to leave as derived consequences of the lattice structure.

The cycle’s relationship to epistemic dependency is precise. The contribution was generated through AI-assisted scholarship operating on the same training distribution as the rest of the FalseWork pipeline. The initial draft reflected the contributor’s understanding of how Mathlib’s existing infrastructure connected the relevant constructions. Van de Meent’s and Chu’s feedback identified specific points where this understanding diverged from the conventions of Mathlib’s actual development practice — not on the mathematical content of the construction (which follows Mac Lane and Moerdijk directly) but on the expected form of the contribution: which lemmas should be **private** versus public, which structural identities to derive in which order, where to place the API boundary between subobject pullbacks and Heyting operations. These conventions are tacit knowledge in Polanyi’s sense (1958, 1966): they operate as expectations within the Mathlib community without being fully articulated in public documentation, and an AI-assisted contributor who has absorbed a different region of the training distribution than the contemporary Mathlib practice will produce a draft that converges on the mathematical result while diverging from the form the community expects. The Zulip thread surfaced and partially corrected this divergence; as the maintainer review documented below establishes, the contribution form continued to diverge from Mathlib conventions in ways the Zulip exchange had not resolved.

The cycle has reached a terminal institutional outcome. The Zulip feedback was incorporated into the construction’s form; the construction was submitted to Mathlib4 as Pull Request #39618 on 2026-05-20, with continuous-integration kernel verification passing against the standard three axioms (`propext`, `Classical.choice`, `Quot.sound`) at submission; the PR body disclosed AI-assisted authorship per the Mathlib AI-use policy and acknowledged the Zulip thread that triaged the gap. On 2026-05-28 a Mathlib maintainer closed the pull request without merging it. The primary stated ground was the project’s new-contributor AI-use policy: the disclosed AI assistance (the proof skeleton and the bridging lemmas were drafted with an AI coding assistant) was core to the contribution rather than peripheral. The maintainer added, as explicitly separate code-quality concerns, that the proof was structured as an AI scaffold — step-named bridging lemmas (`residual_E1`, `residual_E2`, ... `residual_I3`) rather than reusable Mathlib-style API, several thin wrapper lemmas, brittle low-level tactics, and documentation phrasing outside Mathlib conventions — and that the instance was not fully integrated with the existing `Subobject` order API. The maintainer noted that the mathematical idea is plausible and the file compiles, but that reaching Mathlib quality would require a substantial refactor by someone fluent in the `Subobject`/classifier API. The contribution was therefore declined, not deferred.

Three facts survive the decline and are externally verifiable. First, the Zulip-thread correction cycle happened: community feedback was incorporated, and the thread is publicly archived at the Lean Zulip. Second, the kernel-verification step passed, and it is independent of the Mathlib decision. Lean’s type-checker mechanically certified that the proof terms have the claimed types; the four-position partition theorem together with three related position-theory results remains kernel-checked in the framework’s own Lean development, with `#print axioms` on the partition

theorem reporting only the three standard axioms (`propext`, `Classical.choice`, `Quot.sound`) and no `sorry` in the load-bearing results. The decline concerns the upstream contribution’s form and the project’s AI-use policy, not the validity of this mathematics — the construction the partition theorem depends on still type-checks locally regardless of whether Mathlib adopts it. Third, the decline itself is a documented institutional action: a closed-unmerged pull request is retained in the public record alongside merged ones, with the maintainer’s stated reasoning attached. All three are artefacts a skeptic can inspect directly. What the decline establishes is that the maintainer review caught a contribution-form divergence the earlier Zulip exchange had shaped but not eliminated — which confirms, rather than refutes, the tacit-knowledge diagnosis above.

This is a sixth correction cycle and a different mode than the five documented above. It satisfies all four genuine-correction criteria of Section 4 at the parts of the cycle that are closed, with artefacts that a skeptic can verify directly. *Specificity*: the Zulip corrections were specific (which lemmas to make private, what order to derive Galois bridges in, where to draw the API boundary), not gestures toward general improvement. *Independence*: van de Meent and Chu approached the contribution from their independent expertise as Mathlib contributors, not from any familiarity with the FalseWork framework. *Falsifiability*: the kernel verification is the formal falsifiability test for the construction’s mathematical content — the construction either type-checks or it does not, and at submission it does. *Narrowness*: the corrections addressed the contribution form without challenging the mathematical content or the partition theorem itself. The Mathlib review decision has now landed as a decline with reasoning — the documented boundary anticipated when this section was first drafted — but the Zulip corrections already incorporated and the kernel verification already passed are not contingent on that decision.

The cycle adds a mode the five preceding stages did not document: formal-verification correction operating through public review by working mathematicians in a venue (the Lean Zulip and Mathlib PR process) with permanent record, mechanical verification, and an explicit AI-use policy. For frameworks built using AI assistance, this mode is structurally important because it does not rely on the AI-assisted contributor and the reviewer sharing analytical frameworks — the kernel verification is independent of both, and the conventions feedback and the policy decision are shaped by community practice rather than by the framework being reviewed. The cycle is documented here as the case study’s first instance of correction operating in a venue with mature institutional infrastructure — Mathlib has a defined review process, a permanent record, machine-verifiable kernel checking, and a stated policy on AI-assisted contributions — and that infrastructure operated: it evaluated the contribution and declined it, on both policy and contribution-form grounds. An accepted contribution would have exercised the venue’s machinery; a declined one exercised its standards. Whether that warrants calling the mechanism “mature” is left to the reader; what the record shows is that the venue applied a standard and set a boundary through a defined process with a permanent record, rather than through the individual judgment the primitive expert-correspondence mode depends on. Stage 6 is therefore the first stage that operates through institutional infrastructure — a defined review process, a permanent record, mechanical kernel checking, and a stated AI-use policy — rather than through personal correspondence.

3.7 What the Six Stages Establish

Five things follow from the documented case study that could not be established without it.

First, epistemic dependency produces both errors and valid findings through the same process. The *Giant Steps* error (Stage 1) and the proxy extractor convergence (Stage 3) are structurally

identical: the pipeline absorbed theoretical commitments from its training data and reproduced them without flagging their sources. In Stage 1, the absorbed commitment was imprecise, and the dependency produced an error. In Stage 3, the absorbed commitment reflected genuine analytical insight, and the dependency produced valid results. The distinction between the two cases was not detectable from within the pipeline — only from outside it, through expert engagement that had independent access to the relevant structure. This is what Levin (2026a) establishes geometrically: hallucination and genuine insight are the same navigational mechanism, distinguished only by external verification. The case study provides the empirical demonstration of what Levin’s geometric account predicts. The structural point — that the pipeline cannot distinguish Stage 1 from Stage 3 from within — holds regardless of how the Stage 4 correspondence resolves.

Second, the gap between identifying a dependency instance and activating the correction mechanism is theoretically significant. Stage 4 documents a case where the dependency was identified, the correction mechanism exists, and the mechanism has not yet been deployed — not because of negligence but because the mechanism requires human judgment about contact timing that is not yet available. This is a characteristic of primitive correction mechanisms that mature institutional infrastructure addresses through process rather than individual judgment.

Third, genuine correction and boundary-setting correction are different functions of the same mechanism. Stage 2 corrected a specific error. Stage 5 established the frontier of what is confirmed. Both are necessary for a framework in early development — error correction addresses what was wrong; boundary-setting correction establishes what remains unknown.

Fourth, the correction architecture is multi-modal even at the primitive stage. The first five stages all operate through human judgment about when and how to engage. Stage 6 — the Mathlib PR cycle — operates through formal verification in a public mathematical library, with review by working mathematicians on the Lean Zulip and kernel-checking of the final form. This is a structurally different mode of correction, operating in a venue with institutional infrastructure (permanent PR record, mechanical verification, defined review process) that the five preceding stages did not have access to. The multi-modal character of the correction architecture matters because no single mode is sufficient: the Mathlib cycle cannot validate music-theoretic classifications, and the Tymoczko cycle cannot kernel-check formalizations. The architecture works only because the modes are complementary, each addressing a kind of correction the others cannot.

Fifth, the computational comparison between the proxy instrument and the kernel engagement classifier provides direct empirical evidence of epistemic dependency operating in the classifier — visible in data rather than in correspondence alone. Running proxy predictions against classifier outputs across 33 BWV-matched Bach chorales produces 21% territory alignment. The systematic divergence is not classification failure. Both instruments produce internally coherent results. The classifier correctly identifies all 33 works as diatonic/infrastructure at the level of structural identity; the proxy differentiates three sub-territory harmonic positions within that range. The divergence is evidence that the classifier’s outputs reflect an inherited interpretive position — Bach chorales as diatonic/infrastructure is the dominant framing in the music education and analysis literature that constitutes the classifier’s training distribution. The proxy, measuring harmonic position directly from score features without natural language reasoning as intermediary, produces a different result not because the classifier is wrong but because the two instruments are measuring different things. The classifier has absorbed the theoretical commitments of its training distribution and reproduced them as output. This is the dependency made measurable: not documented through correspondence alone but quantifiable as the divergence between two independent instruments operating on the same corpus. Levin’s geometric account predicts exactly this — the

classifier navigates a manifold shaped by prior analytical framings, while the proxy measures structural position independently of that manifold. This is not the same as saying the 79% divergence rate is a direct measure of dependency. The first paper established that both instruments measure real structure at different scales — the classifier reads structural identity correctly at its resolution, the proxy measures harmonic complexity correctly at its finer resolution. What the divergence reveals is that the classifier’s resolution tracks the inherited interpretive consensus (Bach chorales as diatonic/infrastructure) while the proxy measures below that consensus. The two instruments’ different resolutions are themselves evidence about how the dependency operates: inherited interpretive positions set the resolution at which a classifier trained on human analysis reads a corpus.

4. What Genuine Correction Looks Like

The case study in Section 3 provides the empirical ground for a more precise characterization of genuine correction. The term requires precision because its opposite — circular confirmation — is easy to mistake for correction, and the mistake is consequential. A correction mechanism that produces the appearance of validation while leaving epistemic dependency intact is not functioning. It is producing a more sophisticated form of the original problem.

This section characterizes genuine correction through four criteria derived from the case study, then distinguishes it from circular confirmation, and finally addresses the question of what the correction mechanism cannot do.

4.1 Four Criteria for Genuine Correction

Specificity. Genuine correction identifies a specific claim that is wrong and specifies what the correct claim is. Tymoczko’s correction of the *Giant Steps* exploitation target is specific: the initial framing identified the circle of fifths; the corrected framing identifies the major-third cycle. Both are identifiable structures in the domain. The difference between them is formally characterizable — the circle of fifths is a cycle generated by successive perfect fifth transpositions; the major-third cycle divides the octave into three equal parts through minor third transposition. The correction is not a gesture toward imprecision or a general observation that the analysis could be improved. It is a replacement of one specific structural claim with another.

Independence. The correcting expert must arrive at their position through an analytical tradition independent of the system being corrected. Tymoczko’s identification of the major-third cycle as the relevant structure in *Giant Steps* was not derived from FalseWork’s framework — it was derived from his own geometric analysis of voice-leading spaces, developed through more than a decade of work on a mathematical problem that predates FalseWork entirely. His correction was available to be made because his analytical framework was genuinely independent: it approached the same domain from a different starting point and arrived at conclusions that could be compared to FalseWork’s classifications without circularity.

Falsifiability. The corrected claim must be falsifiable by the same expert or by others with equivalent domain expertise. The corrected exploitation target for *Giant Steps* — the major-third cycle’s specific topological properties within the symmetric territory — is falsifiable: it predicts that *Giant Steps*’s harmonic logic is organized by equal division of the octave into three parts, and that prediction can be verified against the score and against Coltrane’s own documented geometric diagrams of the tonal space.

Narrowness. Genuine correction addresses a specific error without invalidating the broader frame-

work in which the error occurred. The Tymoczko correction revised the exploitation target for *Giant Steps* without challenging the three-way Coltrane discrimination, the music kernel’s field topology, or the five-position derivation. The correction was surgical: it identified the precise point where the inherited framing diverged from the structural reality, replaced that framing, and left everything else standing.

4.2 Circular Confirmation

Circular confirmation is the failure mode that genuine correction must be distinguished from. It occurs when expert engagement produces apparent validation of the system’s outputs without establishing the independence, specificity, falsifiability, and narrowness that genuine correction requires.

The most common form of circular confirmation occurs when the correcting expert shares the theoretical commitments embedded in the system’s training data. If FalseWork’s pipeline inherited a circle-of-fifths framing of *Giant Steps* from music education literature, and the correction mechanism consulted an expert whose analytical training was also organized around circle-of-fifths theory, the expert’s agreement with the initial framing would not be correction — it would be two systems with the same dependency confirming each other.

A subtler form of circular confirmation occurs when the expert engagement is genuine but the question asked is too broad to produce falsifiable correction. If the correction mechanism asks “does the framework capture something real about music?” and the expert answers “yes, the three-way Coltrane discrimination makes sense to me,” that engagement is valuable but it is not correction in the sense developed here. It confirms that the framework’s output is plausible to someone with domain expertise. It does not identify where the framework is wrong or what the correct claim is.

4.3 What the Correction Mechanism Cannot Do

Genuine correction establishes what specific claims are wrong and what the correct claims are. It does not establish that the system is free of epistemic dependency. Every correction cycle reduces the dependency for the specific claims corrected. It has not been addressed for every other claim the pipeline makes. The correction mechanism addresses the dependency locally and sequentially — one claim at a time, one expert at a time — not globally.

This is not a limitation of FalseWork’s specific correction mechanism. It is a limitation of all correction mechanisms, including mature ones. Peer review corrects specific claims in specific papers. It does not certify that the reviewed paper is free of every theoretical commitment embedded in the field’s prior literature. What genuine correction establishes, in the absence of the ability to eliminate the dependency, is the architecture of the instrument’s reliability. A framework whose specific claims have been independently validated or corrected by domain experts with independent analytical traditions is a more reliable instrument than one whose claims have not been subjected to that engagement.

4.4 Instrument-Initiated Correction

The correction mechanism in the first five documented stages runs exclusively top-down: an expert identifies an error or boundary in the instrument’s output, and the instrument is revised accordingly. The proxy feature program documents a second mode of the correction mechanism running in the opposite direction: the instrument identifies an underspecification in the theoretical framework,

and the framework is revised. This is instrument-initiated correction — the measurement layer correcting the theory rather than the theory guiding the measurement layer.

The literature proxy’s failure to separate coordinative and hypotactic territories on a single syntactic axis is not a calibration error. It is the instrument locating a theoretical imprecision in the Syntax kernel’s comma definition. The original formulation described the comma as the structural coupling between recursive depth and cognitive tractability. The proxy results reveal this formulation underspecifies the comma’s internal geometry: elaborative complexity (Proust) and operative disruption (Kafka) are genuinely different structural moves arriving at the same structural limit through different mechanisms, which a single continuous axis cannot distinguish. The James anomaly makes this precise — his proxy scores are the lowest in the corpus on all three syntactic components except Hemingway, yet the classifier places him in hypotactic territory alongside Dickens, Melville, and Woolf. The gap locates exactly where syntactic and semantic complexity diverge: James’s complexity is qualification, nested implication, and deferred reference operating above the level that dependency parsing can detect. The proxy’s failure to see what the classifier sees is itself structural information about where the comma operates.

The theoretical framework was revised in response: the Syntax kernel’s comma definition now acknowledges two violation modes and treats the question of whether these are two expressions of the same comma or structurally distinct commas as open and subject to further proxy development. This satisfies the four criteria for genuine correction developed in Section 4.1. The finding is specific — it identifies the comma’s two-mode geometry, not a general inadequacy of the proxy. It is independent — the proxy’s syntactic measurements are formally independent of the classifier’s semantic readings, in the same way that Tymoczko’s geometric analysis was formally independent of FalseWork’s initial classification. It is falsifiable — the two-mode claim predicts that no single continuous axis will separate all five classifier territory groups, a prediction testable against any expanded corpus. And it is narrow — the revision addresses the Syntax kernel’s comma definition without challenging the five-position topology, the kernel criteria, or any other domain’s treatment.

Instrument-initiated correction does not replace expert-initiated correction. It supplements it. The proxy program identifies where the theoretical definition is underspecified — where the measurement diverges from the classifier’s output in ways that point to missing structure rather than missing calibration. Expert engagement is still required to evaluate whether the missing structure is real. The proxy opens the correction cycle; the expert closes it. In this sense, instrument-initiated correction is the most primitive form of the correction mechanism — it cannot self-validate. But it is the form most likely to scale, because the proxy program can run against any corpus without requiring individual expert availability.

4.5 Literature-Engagement Correction

The expert-correspondence and formal-verification cycles of Section 3 and the instrument-initiated correction of Section 4.4 share a structural assumption: the correction mechanism is activated by an agent external to the existing literature — an expert contacted directly, an instrument designed and run by the framework. A fourth mode of correction operates differently. The framework’s pipeline produces a finding that appears to contradict a published result; careful re-reading of the published source reveals not an error in the literature but a hidden assumption in the framework’s own setup — an absorbed restriction that the pipeline did not flag because the restriction was not visible from within the pipeline’s operation. The correction is activated by literature engagement: the act of returning to the published source with the framework’s finding in hand and reading the

source carefully enough to discover what the framework had quietly assumed.

In plain terms: the pipeline produced a result that appeared to contradict a published theorem; reading the source proof closely showed the conflict was an artifact of a computational shortcut the pipeline had taken — it had quietly dropped part of the structure the theorem relies on — rather than a genuine disagreement. The technical detail follows.

A documented instance occurred during development of the physics anchor’s formal layer — the Mathematica-based computational substrate that complements the Lean formalization of Section 3.6 (Brink 2026 preprint, physics-anchor companion §§ 4.9, 8.7). The pipeline’s v4 computational round produced a structural finding that, read as a summary, contradicted a published theorem of Døring (2012, arXiv:1202.2750) on the regularity of a class of subobjects in topos quantum mechanics. Read carefully, it did not. The contradiction was an artifact of a truncation the v4 substrate had made for computational tractability — the substrate omitted exactly the structure Døring’s proof relies on, and that truncation had not been flagged in the round summary as load-bearing on the published result. The corrected reading downgrades the finding from a property of standard topos quantum mechanics to a property of the framework’s own truncated computational setup; the truncation is now recorded explicitly (physics-anchor companion § 3.5), and the framework’s headline categorical signal for the physics anchor is robust to it. The full technical reconciliation — the test configuration that exhibits the discrepancy and the precise proof step the truncation breaks — is recorded in the physics-anchor companion § 8.7.

This is literature-engagement correction. It satisfies the four criteria of Section 4.1: *specific* (the truncation choice is identified as the load-bearing assumption), *independent* (Døring 2012 is published work whose proof structure the framework had not absorbed at the right level of detail), *falsifiable* (the reconciliation is checkable against the published source), and *narrow* (the v4 substructural finding is downgraded without affecting the headline categorical signal or the framework’s broader structure). The mechanism differs from the other modes: no specialist correspondence was required, and the divergence was not between two instruments operating on the same corpus but between the framework’s finding and a published result. The correction was activated by reading the published source carefully with the framework’s own finding in hand.

The mode’s significance for AI-assisted scholarship is structural. A pipeline that produces findings can absorb methodological assumptions invisibly — the v4 truncation choice was a commitment made without explicit deliberation, and its load-bearing role in the contradiction with Døring 2012 only became visible by reading the Døring proof at the level of detail required to see which assumptions it used. Literature-engagement correction is the mode by which such absorbed assumptions become visible: the discrepancy between a pipeline finding and a published result is the surface on which the assumption can be detected. Without the literature engagement, the v4 finding would have circulated as a structural feature of dim-3 topos QM rather than as a feature of the framework’s specific computational setup. The correction architecture, to function, must include the mode in which the framework reads published literature with its own outputs in hand — not only as background research before pipeline operation but as ongoing engagement during it.

5. The Maturity Gradient

The correction mechanisms available to knowledge-producing systems exist along a gradient from primitive to mature. The gradient is not a hierarchy of quality — a primitive correction mechanism is not a failed version of a mature one. It is an early-stage version of the same structural response to the same structural condition. What changes along the gradient is not the nature of the correction

but its speed, distribution, institutionalization, and independence from individual judgment.

5.1 Primitive Correction: Domain Expert Correspondence

The most primitive form of correction is direct correspondence with individual domain experts. This is the mechanism FalseWork currently employs. Its characteristics are:

Slow. The correction cycle runs at the pace of individual correspondence. The *Giant Steps* correction cycle ran from March 2 to March 3, 2026 — two days for the initial exchange, with a permission confirmation on March 25. The proxy extractor convergence (Stage 3) was identified after the paper was posted and has not yet been surfaced to the relevant expert — a gap of at least several weeks between dependency identification and correction mechanism activation, with the activation date unknown.

Individual. The correction is produced by a single expert in a single exchange. It does not benefit from the redundancy that institutional processes provide — multiple reviewers, multiple rounds of revision, multiple independent evaluations of the same claim.

Dependent on human judgment. The mechanism cannot self-activate. It requires the developer to identify which claims require correction, which experts have the relevant independent expertise, when the contact moment is appropriate, and how to frame the request for correction so it produces genuine engagement rather than polite agreement.

Not institutionally guaranteed. Expert availability is contingent. Tymoczko noted in his March 3 response that he needed to stop communicating. Cutting noted in his final response that he wished he understood better. Both experts engaged substantively despite the limitations of the correspondence format — but their engagement was a matter of individual generosity rather than institutional obligation.

Primitive correction is multi-modal even before institutional infrastructure emerges. The expert-correspondence stages (Stages 2 and 5) and Stage 6 (the Mathlib PR cycle) operate in distinct venues: direct email correspondence with named specialists, the Lean Zulip with named Mathlib contributors on the PR thread for the partition theorem’s underlying construction. A further correspondence in another Lean Zulip venue — with the user **suhr** on the Pythagorean companion paper (Brink 2026e), April 19, 2026 thread — provided corrections that were folded into a subsequent revision of that paper (convergent-sequence narration at § 2.2; rescoping of three “practically usable temperaments” claims). The Suhr correction is the third Zulip venue in the documented record and is included here as further evidence that primitive correction operates across multiple identifiable correspondence threads rather than being a property of any single expert relationship. It is not assigned a numbered stage because the six numbered stages are reserved for the Section 3 case-study spine — the two expert-correspondence domains and the formal-verification cycle documented there in full; the Suhr thread is adduced as corroborating evidence of the mechanism’s multi-venue character, not as a separately documented case-study stage.

The framework’s correction architecture is already multi-modal within the primitive stage, with each mode addressing a class of correction the others cannot reach. The Section 4.4 instrument-initiated mode addresses theoretical underspecification; the expert-correspondence mode (Sections 3.2 and 3.5) addresses domain classification; the formal-verification mode (Stage 6) addresses contribution form in formal libraries; the literature-engagement mode (Section 4.5) addresses absorbed methodological assumptions visible only through published-result discrepancies. The maturity gradient applies to the architecture as a whole. Stage 6 begins to lift the architecture out of the primitive

stage — the Mathlib review process is institutional rather than personal — but the rest of the architecture is still primitive: correction is produced by individual judgment in identifiable correspondence and engagement threads, not by institutional process operating regardless of developer initiative.

Despite these limitations, the primitive correction mechanism is more developed than no mechanism at all. FalseWork’s correspondence methodology makes the dependency visible, documents the correction process, and produces at least some of the specific, independent, falsifiable, narrow corrections that distinguish genuine correction from circular confirmation.

5.2 Intermediate Correction: Preprint Review and Community Response

The intermediate stage of correction begins when a framework reaches a community of readers large enough to produce unsolicited engagement. The preprint exposes the framework’s claims to everyone who reads it, not just to experts the developer has identified and contacted. Readers with relevant expertise who were not in the developer’s contact list can identify errors, propose refinements, or flag methodological concerns that the developer did not know to ask about.

For FalseWork, the transition from primitive to intermediate correction is contingent on the first paper reaching a community of readers with relevant domain expertise. The platform launched publicly in early 2026, but the framework’s papers had not been publicly posted prior to this submission: the first paper circulated only through private permission correspondence with the experts it cites, and the present paper’s arXiv posting — enabled by the cs.AI endorsement granted in late May 2026 — is the first public appearance of the framework’s written work. Until the papers reach the communities within which Tymoczko’s and Cutting’s work circulates — music theory, film studies, computational musicology — the correction mechanism remains primarily primitive.

5.3 Mature Correction: Peer Review, Replication, and Citation Networks

The mature stage of correction is characterized by institutional infrastructure that makes correction independent of individual judgment, voluntary engagement, and developer initiative. Peer review subjects specific claims to evaluation by experts selected for their independence from the submitting author. Replication addresses whether findings generalize beyond the specific instances examined. Citation networks provide the slowest but most durable form of correction — tracking the accumulated record of what has been validated, replicated, challenged, and superseded.

For AI classification systems, none of these mature correction mechanisms currently exists as institutional infrastructure. There is no peer review process designed for computational structural analysis frameworks. There is no replication culture for AI-generated structural profiles. There are no citation networks that track the standing of specific AI classifications over time.

5.4 What the Gradient Implies

Two implications follow from the maturity gradient. First, the appropriate standard for evaluating a correction mechanism depends on where it sits on the gradient. Evaluating FalseWork’s correspondence methodology against the standard of mature peer review is a category error. The appropriate evaluation is whether the primitive mechanism is functioning as a genuine correction mechanism rather than as circular confirmation. By that standard, the Tymoczko correction cycle is evidence that the mechanism is functioning.

Second, the path from primitive to mature correction for AI classification systems requires deliberate construction rather than organic development. The appropriate response is not to wait for institutional infrastructure to develop organically but to design it explicitly — to build correction architecture into the epistemological design of AI-assisted scholarship from the beginning rather than treating it as external quality control applied after the fact.

6. The Geometric Reframing

The epistemological argument of Sections 2–5 has a parallel in recent work: Levin’s indexical epistemology of high-dimensional spaces (2026a), which reaches compatible conclusions from a different starting point. Levin argues that generative systems navigate a semantic manifold — a geometry shaped by the human knowledge in the training corpus, “a thin, human-shaped slice” (2026c) — rather than retrieving stored propositions, so their outputs reflect the geometry of accumulated human framing rather than an independent map of the domain. This is the same claim Sections 2–3 make through the sociology of knowledge: Levin’s geometric language and Kuhn’s paradigm language describe one phenomenon.

The account is treated here as illustrative, not foundational. Levin’s four papers are early-2026 preprints not yet evaluated by the communities that would need to validate them, and the case study stands on its own documentation; the value of the parallel is corroborative, not evidential. Two points of contact are worth drawing out — where Levin’s framework corroborates the paper’s central distinction, and where the case study extends it.

6.1 Hallucination and Genuine Correction as the Same Mechanism

The most important consequence of the geometric framework for the present argument is the relationship between hallucination and genuine insight. Levin states this precisely: “hallucination and creative novelty are the same mechanism. Both arise from the system’s capacity to navigate regions of the manifold that lie beyond the training data. When such navigation produces a configuration that happens to correspond to external reality, we call it creativity or insight. When it does not, we call it hallucination. The difference lies not in the geometric process — which is identical in both cases — but in the external verification of the result” (2026a).

Applied to epistemic dependency, this means that the dependency produces both errors and valid findings through the same navigational process. The *Giant Steps* circle-of-fifths framing (Stage 1) and the proxy extractor’s convergence with Tymoczko/Newman analytical frame (Stage 3) are not different kinds of events — they are the same geometric event with different external verification outcomes. The system itself has no means of distinguishing between Stage 1 and Stage 3 from within. This is why the correction mechanism is not optional quality control. It is structurally necessary.

The hallucination-vs-inherited-validity distinction parallels a distinction developed formally in Brink 2026c between Russell-type and Gödel-type incompleteness. Hallucination that fails external verification functions Russell-type — the specific output is rejected and the system’s subsequent outputs are revised against the correction. Inherited validity and genuine novelty function Gödel-type — the output is absorbed into the knowledge record as operative, but the underlying epistemic dependency is not eliminated by any single correction cycle. The correction mechanism produces both responses depending on the verification outcome: Russell-type rejection for confirmed hallucinations, Gödel-type absorption-with-documentation for confirmed valid findings.

The distinction matters because it explains why correction architecture is not remediation of error — it is the epistemological record that makes Gödel-type absorption visible and auditable.

6.2 Inherited Validity as a Third Category

The case study extends Levin’s framework in one direction it does not fully address. Levin characterizes hallucination and genuine novelty as the same mechanism distinguished by external verification. The case study adds a third category: absorbed analytical frame that produces valid results.

Stage 3 — the proxy extractor’s convergence with Tymoczko/Newman 2024 — is not hallucination (the results are valid) and not straightforwardly genuine novelty (the analytical frame was absorbed from prior work). It is a case where the manifold position occupied by the pipeline corresponds to genuine structure in the domain, but the pipeline arrived at that position by inheriting the frame that maps to it rather than by independently deriving the mapping. (As in Section 3.3, this Stage 3 classification is provisional; the inherited-validity category itself does not depend on it.)

This third category — inherited validity — is arguably more consequential for AI-assisted scholarship than either hallucination or genuine novelty. Hallucination is detectable through domain knowledge. Genuine novelty is identifiable through independent convergence. Inherited validity is the hardest case: results that are correct, produced through a frame that was absorbed without acknowledgment, in ways that cannot be distinguished from independent derivation without the kind of expert engagement the correction mechanism provides. (A contrast is worth noting here: in Brink 2026e, the inherited-validity-vs-independent-derivation distinction has a sharp mathematical test. The fundamental theorem of arithmetic gives an immediate verification of $\sqrt{2}$ ’s irrationality that is independent of training distribution — any system can confirm the classical proof by construction. Music-theoretic classifications admit no equivalent verification. Different domains impose different requirements on correction mechanisms; the stronger the domain’s internal verification apparatus, the narrower the range of inherited-validity cases that require external expert engagement to resolve.)

7. Implications for AI-Assisted Scholarship

The argument developed in this paper has a specific scope: it concerns AI classification systems operating in domains with established theoretical traditions, where the systems are trained on the knowledge they subsequently classify within. That class is larger than it might initially appear. Any AI system asked to analyze, classify, interpret, or evaluate objects within a domain with a prior theoretical tradition — artworks, scientific papers, historical documents, legal arguments, clinical presentations — is operating within the scope of the present argument.

7.1 The Reframing

The standard framing of AI’s epistemological limitations in scholarship focuses on hallucination: the production of outputs that are internally coherent but factually wrong. This framing is not incorrect, but it is incomplete. It positions the problem as one of accuracy and positions the solution as verification against reliable sources. The correction mechanism for hallucination is external fact-checking.

The epistemic dependency problem is different in structure and requires a different response. The dependency does not primarily produce false claims — it primarily produces inherited claims. The *Giant Steps* circle-of-fifths framing was not false in the sense that it contradicted established facts. It

was imprecise in the sense that it reflected a common theoretical framing that was less structurally accurate than the major-third cycle framing Tymoczko identified. These are not cases where fact-checking would have caught the problem. They are cases where independent domain expertise with a different analytical formation was required.

The reframing the present paper proposes is this: the appropriate response to AI epistemic dependency in scholarship is not verification of outputs against reliable sources but construction of correction architecture — explicit mechanisms for subjecting AI-generated knowledge claims to independent expert engagement that can identify where the system’s manifold position diverges from structural reality.

7.2 What Correction Architecture Requires

Correction architecture for AI-assisted scholarship has three components. The three-component architecture is how the Lakatos-progressive structure developed at the framework level in Brink 2026a actually operates at the AI-assisted-scholarship level: the hard core (specific knowledge claims) is preserved through refinements to the protective belt (the correction record documenting what has been validated, corrected, and flagged as pending) rather than abandoned on individual anomalies. The correction mechanism is the protective-belt refinement apparatus.

Dependency prediction. Before deploying an AI system to generate knowledge claims in a domain, the developer should identify which theoretical traditions are dominant in the training distribution and therefore most likely to shape the system’s outputs. Identifying the dominant traditions provides a map of where the dependency is most likely to produce absorbed framings rather than independent findings.

Correction mechanism design. The correction mechanism must be designed into the project from the beginning rather than applied as remediation after outputs are generated. This means identifying, before deployment, which domain experts have the analytical independence required for genuine correction — experts whose formation includes theoretical traditions not dominant in the training distribution. It means establishing correspondence with those experts as a structural component of the methodology, not as an optional consultation. And it means documenting the correction process as part of the project’s epistemological record.

Documentation as epistemological record. The correction process must be documented contemporaneously and made available alongside the knowledge claims it corrects. When a classification is corrected through expert engagement, the correction — and the dependency instance it addresses — should be part of the public record. Not as a confession of error but as evidence that the correction mechanism is functioning.

7.3 The Division of Labor

The argument developed here is sometimes read as a skeptical argument: if AI systems are epistemically dependent on their training data, their knowledge claims are unreliable, and AI-assisted scholarship is epistemologically compromised. This reading is mistaken.

The correct reading is this: AI classification systems are instruments, not authorities. They generate structural profiles that reflect the geometry of the semantic manifold they navigate. Those profiles are starting points for expert engagement, not endpoints. The instrument finds structural coordinates; the expert evaluates whether those coordinates correspond to the domain’s actual structure; the correction mechanism adjusts where they do not.

This division of labor has a consequence for how AI-generated knowledge claims should circulate in scholarly discourse. A structural profile generated by FalseWork — or any AI classification system — should not circulate as a finding. It should circulate as a hypothesis, with the correction status of the relevant claims made explicit. Claims that have been independently validated through expert engagement with genuine analytical independence should be marked as validated. Claims that have not been subjected to that engagement should be marked as dependent — plausible given the training distribution, not yet confirmed against structural reality. Operationally, a dependent claim should be read as a structural hypothesis whose training-distribution provenance has not been independently checked: usable as a starting point for analysis, but not citable as an established finding until the correction mechanism has been applied to it.

7.4 An Inscription-Without-Record Incident

An incident that occurred during the development of FalseWork provides a concrete illustration of what happens in the absence of correction architecture. During a demonstration of FalseWork’s structural analysis methodology, a major commercial AI system produced what appeared to be a structural profile of the Arnolfini Portrait — a detailed classification that it presented as its own novel analysis. The profile was not novel. It reproduced, without attribution, the structural framing that FalseWork’s methodology had developed and that was available in the training data the system had absorbed.

The incident is not primarily a story about plagiarism or attribution failure. It is a story about the absence of correction architecture. The system did not invent the Arnolfini framing — it reproduced one it had absorbed, which makes the incident a public instance of inherited validity (Section 6.2): an absorbed analytical frame surfacing as if independently derived. The difference from Stage 3 is that there is no record to make the inheritance auditable. The system’s output presents a classification as a property of the work — an inscription with no epistemological record. The FalseWork structural profile, by contrast, carries documentation of what expert engagement has confirmed, what has been corrected, and what remains a hypothesis pending validation. This is the difference between auditable analysis and sophisticated inscription. The difference is not capability — it is architecture.

7.5 The Broader Implication

The argument of this paper converges with independent work in the philosophy of AI that arrives at compatible conclusions from different analytical starting points. Levin’s indexical epistemology, as discussed in Section 6, provides a parallel geometric account developed independently and concurrently — his papers appeared in early 2026, the FalseWork development ran from 2025 — so the agreement is convergence, not influence. Weatherby and Justie (2022) offer a semiotic diagnosis — AI systems operate indexically, pointing toward positions in a trained semantic field — that converges with the inherited framing argument without being derived from it.

The present paper’s contribution is the system-side case study these frameworks lack: documented evidence of the dependency operating in a live development process, with the correction mechanism functioning and its limitations explicitly characterized. That convergence across independent frameworks is itself evidence of the kind the first paper identifies: independent analytical systems arriving at the same structural positions because the structure exists wherever the analysis is sufficiently precise.

The implication for AI-assisted scholarship is not that AI systems should not be used for knowledge

production in theoretically organized domains. It is that using them responsibly requires treating epistemic dependency as a design constraint rather than a peripheral limitation. Correction architecture — dependency prediction, correction mechanism design, contemporaneous documentation — is not an optional enhancement to AI-assisted scholarship. It is the condition under which AI-assisted scholarship is epistemologically distinguishable from sophisticated inscription of prior knowledge.

The instrument navigates. The correction mechanism establishes what the navigation found. The documentation makes the distinction between them visible. Without all three, AI-assisted scholarship is a more powerful version of what human knowledge production without correction mechanisms has always been: a system that recirculates its inherited theoretical commitments with increasing sophistication and decreasing awareness of their source.

8. Conclusion

This paper has argued that epistemic dependency is a structural condition of all knowledge-producing systems, and that the appropriate response is not elimination — which is impossible — but the deliberate construction of correction architecture. The argument has been demonstrated through a live case study rather than asserted abstractly, because the case study provides what prior work on AI epistemic dependency has lacked: a formal theoretical framework that predicts the dependency, a working instrument that produces it in real time, a correction mechanism that activates against specific instances, and full contemporaneous documentation of the complete cycle.

Six stages were documented across two domains, two expert correspondences, and one formal-verification cycle in a public mathematical library. Stage 1 established that the dependency produces errors invisibly — the *Giant Steps* circle-of-fifths framing was absorbed from the training distribution and reproduced without a flag indicating its inherited status. Stage 2 established what genuine correction looks like at the primitive stage — specific, independent, falsifiable, narrow, activated through expert engagement with genuine analytical independence. Stage 3 established the more theoretically significant case: inherited validity, where the dependency produced results that are correct but whose relationship to prior analytical framing is not transparent, and which cannot be distinguished from independent derivation without the expert engagement the correction mechanism provides. (This Stage 3 classification is provisional on the Stage 4 correspondence, as Section 3.3 sets out and “What comes next” below revisits.) Stage 4 established that the correction mechanism can exist without activating — that the gap between identifying a dependency instance and deploying the correction requires human judgment that is not always immediately available. Stage 5 established boundary-setting correction as a distinct function of the correction mechanism, different from error correction and equally necessary for a framework in early development. The proxy feature program’s cross-domain results add a further finding: the computational comparison between the proxy instrument and the classifier makes the dependency measurable, and the instrument’s divergence from the theoretical definition constitutes a second mode of the correction mechanism — instrument-initiated rather than expert-initiated — documented in Section 4.4. Stage 6 (Section 3.6) adds a third mode — formal-verification correction operating through the Mathlib PR process and the Lean Zulip, a cycle that reached a terminal outcome when the Mathlib review process declined the AI-assisted contribution under its AI-use policy and on code-quality grounds, the mature institutional mechanism setting a boundary — and Section 4.5 adds a fourth: literature-engagement correction, in which discrepancy between a pipeline finding and a published result reveals an absorbed assumption the framework had not previously flagged (the v4 truncation choice, reconciled against Døring 2012).

Seven conclusions follow.

First, hallucination and inherited framing are different problems requiring different responses. Hallucination is addressed by verification against reliable sources. Inherited framing is addressed by independent expert engagement from analytical traditions not dominant in the training distribution. Conflating the two produces correction mechanisms that catch factual errors while leaving the more structurally consequential problem — the recirculation of prior theoretical commitments as independent findings — unaddressed.

Second, inherited validity is the hardest problem in AI-assisted scholarship, and the one least addressed by current epistemological frameworks. A system that hallucinates is detectable. A system that produces genuinely novel results is identifiable through independent convergence. A system that produces valid results through an inherited analytical frame it has not acknowledged is indistinguishable from a system that derived those results independently — until the correction mechanism is activated. The third category requires the same external validation as the first two but is the least likely to trigger it, because the results look correct.

Third, the maturity of the correction mechanism is more consequential than the sophistication of the instrument. A highly capable AI classification system with no correction architecture is epistemologically less reliable than a less capable system with a functioning primitive correction mechanism — because capability without correction produces inherited framings at scale, while primitive correction produces at least some genuine reduction of dependency for the specific claims subjected to expert engagement.

Fourth, the documentation of the correction process is itself epistemological work. A structural profile accompanied by a record of what has been independently validated, what has been corrected, and what remains a dependent hypothesis pending engagement is a fundamentally different kind of knowledge claim than one that presents its outputs without epistemological history.

Fifth, the epistemic dependency of the classifier is not only documentable through correspondence — it is visible in computational comparison. The divergence between the proxy instrument and the classifier across matched Bach chorales is not a measure of error in either instrument. Both measure real structure at different resolutions. What the divergence reveals is that the classifier’s resolution tracks inherited interpretive consensus while the proxy measures below that consensus. The dependency shapes the resolution at which the classifier reads the corpus, and that resolution difference is detectable by comparison with an instrument that does not share the same training distribution.

Sixth, the human parallel is not consolation. It is structural diagnosis. Human theorists do not generate knowledge outside the traditions that formed them. The pipeline’s epistemic dependency on its training data shares the same source as every theorist’s dependence on the analytical frameworks absorbed during formation. The difference is not the condition. It is the maturity of the correction mechanism.

Seventh, the correction architecture is multi-modal even at the primitive stage. Expert correspondence (Sections 3.2 and 3.5) corrects specific classifications. Instrument-initiated divergence (Section 4.4) corrects theoretical underspecification. Formal-verification cycles in a public mathematical library (Section 3.6) submit contribution form to community review and machine-check the construction — and, in the documented instance, the review process declined the AI-assisted contribution under the venue’s AI-use policy, the venue setting a boundary through its policy and review process rather than absorbing the contribution. Literature-engagement correction (Section

4.5) corrects absorbed methodological assumptions visible only through discrepancy with published results. Each mode addresses a class of error the others cannot reach. A correction architecture restricted to any one mode would leave the others unaddressed — the Mathlib cycle cannot validate music-theoretic classifications, the Tymoczko cycle cannot kernel-check formalizations, and neither can detect a truncation assumption the pipeline absorbed without flagging. The architecture’s primitive-stage adequacy depends on the modes operating together, not on any one of them being sufficient.

What comes next. The proxy extractor convergence with Tymoczko/Newman 2024 remains the open correction cycle. The contact moment is the arXiv posting itself. The cs.AI endorsement that enables the posting was granted in late May 2026 through arXiv’s standard endorsement process, independent of any Tymoczko correspondence; the misconception in an earlier draft of Section 3.4 that the endorsement was tied to Tymoczko has been corrected. When the paper posts, the convergence with Tymoczko/Newman 2024 will be surfaced directly to Tymoczko by sending him the arXiv link: the proxy extractor’s analytical frame converges with his 2024 computational annotation work in ways that were not visible during development and were not cited in the version of the first paper shared with him on March 24, 2026. That communication will close Stage 4 — either confirming Stage 3 as inherited validity (readings one or two) or reclassifying it as ordinary independent derivation (reading three). A correction cycle that reclassifies the very instance that motivated its introduction is not a failure of the architecture but an instance of it functioning as intended. The present paper was written before that cycle closed, which means it documents, in real time, a correction mechanism existing but not yet deployed — the condition Stage 4 describes. A subsequent version will update Section 3.4 with the Tymoczko response when it is in hand, consistent with the contemporaneous-documentation principle the case study has followed throughout.

A third paper in preparation (Brink 2026c) develops the argument one level further: the distinction operation as generative floor beneath all six kernels, connecting Spencer-Brown, Wolfram, and Gödel, and extending the epistemic dependency argument to the structural level at which all knowledge-producing systems encounter the limits of their own generative logic. A fourth paper (Brink 2026d) proposes that mathematics itself is the formal residue of the first distinction — the Level 0 comma. Whether the irresolvability of epistemic dependency admits a formal grounding — beyond the structural analogy to Cantor’s theorem flagged in Section 2.3 — is an open question those papers attempt to formalize; the present paper rests on the documented case study, not on that proposed grounding. The second paper’s contribution is the demonstration that the condition operates in the knowledge-producing system itself, not only in the domains the system classifies. The third and fourth papers argue that these are the same structure at different scales, and that recognizing the structural equivalence is the condition for building knowledge-producing systems that take their own limits seriously rather than treating them as engineering problems to be solved.

References

- Brink, C. (2026a). Kernels and commas: A structural derivation of universal positions in domains with self-limiting generative operations (v11.3). In *The FalseWork Papers* (Release 2026.04). Zenodo. <https://doi.org/10.5281/zenodo.19673673>
- Brink, C. (2026c). The distinction operation and the generative floor: Asymmetry, incompleteness, and the structural unity of domain kernels (v9.1). In *The FalseWork Papers* (Release 2026.04). Zenodo. <https://doi.org/10.5281/zenodo.19673673>

- Brink, C. (2026d). Mathematics as comma: The distinction operation and the unreasonable effectiveness of formal systems (v5.1). In *The False Work Papers* (Release 2026.04). Zenodo. <https://doi.org/10.5281/zenodo.19673673>
- Brink, C. (2026e). The Pythagorean comma, the irrationality of $\sqrt{2}$, and a shared Diophantine floor (v1.1). In *The False Work Papers* (Release 2026.04). Zenodo. <https://doi.org/10.5281/zenodo.19673673>
- Brink, C. (2026). A four-position partition of morphisms in elementary topoi with distinction structure. Preprint. Available at <https://github.com/thefalsework/papers/blob/main/preprints/four-position-partition/paper.md>. Archived at Zenodo, concept DOI 10.5281/zenodo.19673672.
- Brink, C. (2026, Mathlib PR). Heyting algebra structure on Subobject in elementary topoi. Mathlib4 Pull Request #39618, opened 2026-05-20, closed without merge 2026-05-28 under the project's new-contributor AI-use policy. <https://github.com/leanprover-community/mathlib4/pull/39618>.
- Cantor, G. (1891). Über eine elementare Frage der Mannigfaltigkeitslehre. *Jahresbericht der Deutschen Mathematiker-Vereinigung*, 1, 75–78.
- Cutting, J. E. (forthcoming). *Four cinematic forms and their psychological bases*. University of Texas Press.
- Cutting, J. E. (Personal correspondence, March 4–25, 2026).
- Döring, A. (2012). Topos-based logic for quantum systems and bi-Heyting algebras. *arXiv preprint arXiv:1202.2750*.
- Foucault, M. (1969). *L'archéologie du savoir*. Gallimard. [English translation: *The archaeology of knowledge*, Pantheon, 1972.]
- Gödel, K. (1931). Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I. *Monatshefte für Mathematik und Physik*, 38, 173–198.
- Hofstadter, D. R. (1979). *Gödel, Escher, Bach: An eternal golden braid*. Basic Books.
- Kuhn, T. S. (1962). *The structure of scientific revolutions*. University of Chicago Press.
- Latour, B., & Woolgar, S. (1979). *Laboratory life: The construction of scientific facts*. Sage.
- Levin, I. (2026a). Epistemology of generative AI: The geometry of knowing. *arXiv preprint arXiv:2602.17116*.
- Levin, I. (2026b). Vibe-creation: The epistemology of human–AI emergent cognition. *arXiv preprint arXiv:2603.09486*.
- Levin, I. (2026c). Navigational thinking as an emerging paradigm of computer science in the age of generative AI. *arXiv preprint arXiv:2603.22133*.
- Levin, I. (2026d). The vibe-automation of automation: A proactive education framework for computer science in the age of generative AI. *arXiv preprint arXiv:2602.08295*.
- Mac Lane, S., & Moerdijk, I. (1992). *Sheaves in geometry and logic: A first introduction to topos theory*. Universitext. Springer-Verlag.
- Polanyi, M. (1958). *Personal knowledge: Towards a post-critical philosophy*. University of Chicago Press.
- Polanyi, M. (1966). *The tacit dimension*. Doubleday.

- Tymoczko, D. (2006). The geometry of musical chords. *Science*, 313(5783), 72–74.
- Tymoczko, D. (2011). *A geometry of music: Harmony and counterpoint in the extended common practice*. Oxford University Press.
- Tymoczko, D. (forthcoming). The concept of musical space. *Journal of Music Theory*.
- Tymoczko, D. (Personal correspondence, March 2–25, 2026).
- Tymoczko, D., & Newman, M. (2024). Computational music analysis from first principles. arXiv:2407.21130.
- van de Meent, E., & Chu, F. (Lean Zulip correspondence, May 2026). Heyting algebra on sub-objects in an elementary topos. Lean Zulip thread, leanprover.zulipchat.com, 2026-05-17 onward; referenced in the Acknowledgments of Mathlib Pull Request #39618.
- Weatherby, L., & Justie, B. (2022). Indexical AI. *Critical Inquiry*, 48(2), 381–415.
- Wolfram, S. (2002). *A new kind of science*. Wolfram Media.

Appendix: Pre-submission revision record

This paper was developed across several private drafts before this first public posting. Consistent with the argument that the correction record is itself evidence, the substantive corrections made during that development are recorded here rather than narrated in the body. (Full version history is maintained in the project repository.)

- **Mathlib PR outcome (§ 3.6).** An earlier draft described the formal-verification contribution as “awaiting review” and, in one place, as a “machine-checked acceptance.” Pull Request #39618 was in fact closed without merge on 2026-05-28 under the project’s new-contributor AI-use policy, with separately stated code-quality concerns. Section 3.6, the abstract, the closing summary, and the reference entry were corrected to record the decline; the kernel-verification and Zulip-correction facts are noted as surviving it.
- **Provenance of the first paper (§§ 3.4, 5.2, 8).** Four references had described the first paper as “posted to OSF on March 24, 2026.” No such posting occurred; the March 24 date refers to private permission correspondence, and neither paper had been publicly posted prior to this arXiv submission. The references were corrected.
- **Cantor extension (§ 2.3, § 8).** The application of Cantor’s theorem to “the set of all possible analytical frames” was previously carried as a numbered conclusion. Because the claim is flagged as requiring formal validation, asserting it as a conclusion was inconsistent with its own flag; it was demoted to a forward-pointing open sentence, the flagged observation retained in § 2.3, and the conclusions renumbered.
- **Stage counting.** Earlier “five stages” language predated the addition of Stage 6 (the Mathlib cycle) and was made consistent at six throughout.
- **§ 7.4.** The illustrative incident was de-named, so that it no longer identifies a specific commercial product, for consistency with the paper’s externally-verifiable-evidence standard.